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Machine Learning-Based Prediction of Wax-Related Production Losses for Flow Assurance Optimization

Javad Iskandarov, Ramanzani Kalule, Ravan A. Farmanov, Mursal Zeynalli, and Strahinja Markovic, Chemical and Petroleum Engineering Department, Khalifa University of Science and Technology, Abu Dhabi, UAE; Emad W. Al-Shalabi, Chemical and Petroleum Engineering Department, Khalifa University of Science and Technology, Abu Dhabi, UAE / Research and Innovation Center on CO2 and Hydrogen, Khalifa University of Science and Technology, Abu Dhabi, UAE; Hamid Abderrahmane, Mechanical and Nuclear Engineering Department, Khalifa University of Science and Technology, Abu Dhabi, UAE; Omar Al-Farisi, Dragon Oil, Dubai, UAE

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Abstract

Wax deposition poses a significant challenge to flow assurance in oil production systems, potentially leading to production losses and increased maintenance requirements. Accurately forecasting the impacts of wax deposition on hydrocarbon production can enhance the mitigation strategies and optimize field operations. This study developed supervised machine learning (ML) models to predict oil, water, and gas production rates, using key wellhead, surface, and wax-related parameters. Both tree-based and neural network algorithms were used. A dataset containing 306 samples from multiple wells was used, with 80% utilized for training and 20% reserved for testing the models. Ensemble tree-based models delivered the most accurate performance, with Extra Trees achieving an R^2 above 0.98 for all targets. The performance of the deep learning model, however, was limited by the small dataset size. Singular Value Decomposition analysis was performed to understand how different parameters influence well production dynamics. The results indicated that the wax accumulation rate and wax cutting frequency are essential contributors to well dynamics, alongside tubing head pressure and choke size. Wax-related features were further explored through sensitivity analysis, which showed that increased wax deposition and intervention frequency resulted in reductions of up to 25% in oil rates and 31% in gas rates. Based on the predictions, oil production was more negatively affected by the frequency of wax cutting, whereas the wax accumulation rate had a more severe impact on gas production. Therefore, careful design of flow assurance measures, balancing reactive strategies, such as mechanical wax cutting, with proactive measures, such as chemical inhibition, is critical. The proposed ML workflow provides a reliable predictive framework for hydrocarbon production to support flow assurance decisions, reducing wax-related downtime and enhancing field productivity.

Introduction and Background

Oil production is a complex process that requires the management of multiple wells, surface facilities, and interconnected pipelines to ensure the continuous fluid extraction. The process occurs across interconnected platforms, where oil, gas, and other fluids produced from multiple wells are routed through a system of risers, manifolds, separators, desanders, and pumps. Maintaining the production efficiency in such an environment requires careful monitoring of well performance, pressure fluctuations, and fluid composition to ensure optimal operations. Production challenges, such as pressure drops, high water cut, and various deposits in upstream and downstream facilities, can significantly impact overall recovery.

Accumulation of both inorganic and organic deposits, such as scale, wax, and asphaltenes, is among the key flow assurance challenges in oil production. They can severely impact the hydrocarbon production by reducing flow efficiency and increasing the operational expenditures. These deposits can form and accumulate in wellbore tubing, flowlines, and surface equipment, leading to pressure drops, reduced production rates, and even complete blockages. Scale deposition occurs when dissolved minerals precipitate in the produced water, often triggered by changes in temperature, pressure, pH, or the mixing of incompatible brines (Kamal et al., 2018). Some of the common scale types found in oil wells are calcium carbonate ($CaCO_3$), the most prevalent carbonate scale, particularly in topside facilities and the upper sections of production tubing (Lakshmi et al., 2013). Elevated pH due to excess carbonate ions and pressure drops during fluid expansion can further promote the scale precipitation (Ramstad et al., 2005). Moreover, High-temperature environments are more prone to scale deposition due to reduced solubility of $CaCO_3$ (Hamid et al., 2016). Other problematic scales include barium sulfate ($BaSO_4$) and strontium sulfate ($SrSO_4$), which form when seawater rich in sulfates comes into contact with barium- or strontium-rich formation water, resulting in hard, insoluble scales. Additionally, iron sulfide (FeS) can form due to H_2S corrosion and reaction with iron-containing materials. This may result in a reduction in tubing diameter, increased frictional losses in the wellbore, and damage to downhole equipment. Removal of FeS can be achieved through the use of costly chemical dissolvers, scale inhibitors, or mechanical descaling (Nasr-El-Din and Al-Humaidan, 2001).

Asphaltenes are heavy, polyaromatic components of crude oil that exist in a colloidally dispersed state under reservoir conditions. They become a major flow assurance problem because changes in operating conditions can destabilize this equilibrium, causing precipitation and deposition along the production system. Asphaltene precipitation occurs when reservoir pressure decreases below the asphaltene onset pressure (AOP), during gas injection that reduces the stabilizing resin concentration, or when environmental changes decrease solubility in the wellbore (Leontaritis and Mansoori, 1987). Once precipitated, asphaltene particles tend to aggregate and deposit on subsurface and surface facilities, restricting the flow and enhancing the pressure losses. In extreme instances, deposition may even clog the flowlines and cease the production operation, requiring costly remediation (Buckley, 2012). The extent of deposition depends on the composition of crude oil, pressure changes, and production strategies such as enhanced oil recovery, which can significantly increase asphaltene instability. The common mitigation methods include injection of chemical inhibitors, solvent treatments to dissolve deposits, and mechanical cutting when plugging becomes severe (Stratiev *et al*., 2025).

Wax deposition, which is the primary focus of this study, presents a similarly critical flow assurance challenge, particularly in wells producing high-wax-content oil. Petroleum waxes are primarily composed of high-molecular-weight paraffinic hydrocarbons, including n-alkanes, iso-alkanes, and cycloalkanes, which can crystallize out of solution when conditions change. When the temperature of produced fluids falls below the wax appearance temperature (WAT), these molecules begin to solidify and adhere to tubing and pipeline walls, progressively restricting the flow path. The severity of deposition is influenced by temperature drops along the flow path, crude oil composition, and low flow velocity, which allows wax particles to settle and accumulate (Aiyejina *et al*., 2011). Over time, wax buildup increases pressure losses and, if left unmanaged, can lead to complete blockages. Wax buildup is typically prevented by injecting chemical inhibitors that inhibit crystal adhesion, as well as mechanical interventions such as slickline wax cutting or pigging.

In addition to deposition-related challenges, some wells require artificial lift to sustain production as reservoir pressure declines. Gas lift is commonly applied in oil prdocution operations due to its reliability, flexibility, and ability to operate under varying fluid conditions. It works by injecting high-pressure gas into the production tubing, reducing the density of the fluid column and lowering the bottomhole flowing pressure, so that hydrocarbons can flow to the surface even when natural drive is insufficient (Khamehchi and Mahdiani, 2017). Its advantages include the ability to handle high gas–liquid ratios, continued operation despite increasing water cut, and minimal downhole mechanical components, which reduces the risk of failure (Brown, 1982; BenAmara, 2016). However, its performance depends on proper injection control and a reliable gas supply; the system can be affected by deposits in the injection hardware, requiring preventive maintenance (Nguyen, 2020).

Predicting well performance, where wax deposition, intervention frequency, operational well parameters, and artificial lift performance interact, is challenging to achieve accurately using traditional methods (Nsiama *et al*., 2024). Another significant challenge arises from the sparsity of available data, a common situation in oil production, where interpretations to fill the data gaps render the results less reliable (Mkono *et al*., 2025). Integrating multiple datasets, such as surface facility data, operational parameters, and time, can enhance overall coverage; however, this process also introduces additional sources of uncertainty that propagate through the analysis (Khan *et al*., 2024; Ma *et al*., 2025). Machine learning models enable the combination of diverse operational and reservoir datasets, capturing complex nonlinear relationships, and producing more accurate predictions of production behavior. By learning from historical data, ML can improve the understanding of dynamic reservoir processes and enhance predictive accuracy (Liu and Pyrcz, 2023; Nagao *et al*., 2024). These capabilities have already been demonstrated in applications for production forecasting (Fan *et al*., 2025) and predicting wax deposition trends and their operational impacts in producing wells (Zhang *et al*., 2023), providing a strong basis for applying similar methods to quantify the effect of wax accumulation and intervention strategies on oil and gas production rates.

This study aimed to develop a machine learning framework for predicting oil, gas, and water rates from an offshore platform by utilizing wellhead and wax-related operational parameters as input features. Singular Value Decomposition (SVD) was employed to identify the parameters that most significantly influence dynamic production behavior. The trained model was then used to analyze the impact of wax accumulation and mechanical interventions on production rates in both high- and low-pressure wells.

Methodology

The dataset used in this study comprises 306 data points collected from nine high-pressure and four low-pressure wells. Among those points, 271 are from low-pressure wells and 35 are from high-pressure wells. Each data point represents seven unique system and operational parameters, either directly measured or derived from the field. The distribution of these input and target variables (oil, water, and gas production rates) is illustrated in Figure 1. High-pressure wells in the dataset are defined as those with tubing head pressure (THP) greater than 30 bar, while low-pressure wells have THP values between 9 and 30 bar. Furthermore, according to the data, high-pressure wells exhibit more severe wax deposition, with approximately twice the wax accumulation rate and 50% higher wax cutting frequency compared to low-pressure wells. A brief description of each parameter can be found below:

- **Choke:** Represents the choke size at the wellhead.
- **Tubing Head Pressure (THP):** The pressure at the top of the tubing, reflecting the upstream condition of the well before choke.

- **Pressure Downstream of Choke (Pressure DS):** Indicates the pressure immediately after the choke.
- **Temperature Downstream of Choke (Temperature DS):** Refers to the fluid temperature after passing through the choke.
- **Gas Lift Injected:** The volume of gas injected for artificial lift operations per day, which helps maintain flow in wells with insufficient reservoir pressure.
- **Wax Cutting Frequency (WCF)**: The frequency of mechanical wax removal events, expressed in cuts per hour.

$$\mathrm{WCF} = \frac{1}{\text{Number of hours between two wax cuts}} \tag{1}$$

- **Wax Accumulation Rate (WAR):** The rate at which wax deposits form along the tubing, expressed in meters per hour.

$$\mathrm{WAR} = \frac{\text{Length of wax covered tube}}{\text{Number of hours between two wax cuts}} \tag{2}$$

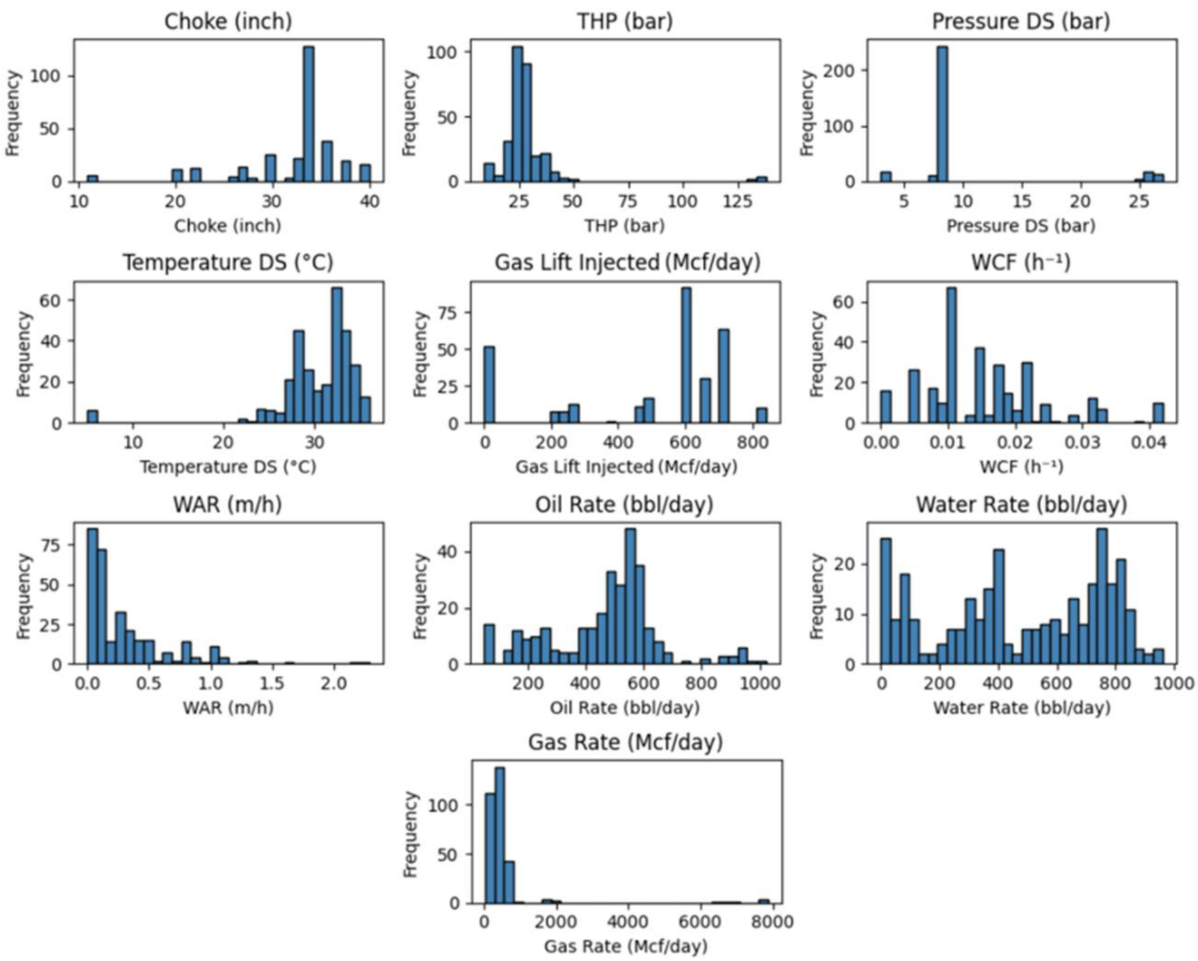

Figure 1—Distribution of input and target variables.

ML algorithms were deployed in this study to predict oil, water, and gas production rates using wellhead properties and wax-related operational parameters as inputs. The input features included Choke (inch), THP (bar), Pressure DS (bar), Temperature DS (°C), Gas Lift Injected (Mcf/day), WAR (m/h), and WCF (h^{-1}). The target output variables were Oil Rate (bbl/day), Water Rate (bbl/day), and Gas Rate (Mcf/day). To identify the most accurate model, a comparison based on performance on the test set was conducted across several widely recognized supervised ML algorithms. Following this, the selected optimized model was used to make predictions for both low- and high-pressure wells, assessing the impact of wax-related features on production behavior.

The evaluated algorithms include Linear Regression (LR), Decision Tree (DT), Gradient Boosting (GB), Random Forest (RF), Extra Trees Regressor (ERT), CatBoost, and a deep learning model: the Multilayer Perceptron (MLP). The combined dataset from all 13 wells was initially split into 80% for training and 20% for blind testing. A 5-fold cross-validation was further applied to the training data. The accuracy of the algorithms was measured based on three metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2).

Results and Discussion

The SVD analysis was applied separately for low- and high-pressure wells to assess how the input features contributed to the production dynamics. The primary objective was to determine the number of principal components or orthogonal modes necessary to capture the variability of the inputs in both cases and identify the dominant contributors to each mode. The results showed a significant difference in both pressure regimes, as shown in Figure 2. For the high-pressure wells, only four principal components were required to cover almost entirely the input variance, with two components accounting for more than 75%. In comparison, low pressure exhibited more complex feature interactions, requiring six modes to describe the variance fully and three for the 75% threshold.

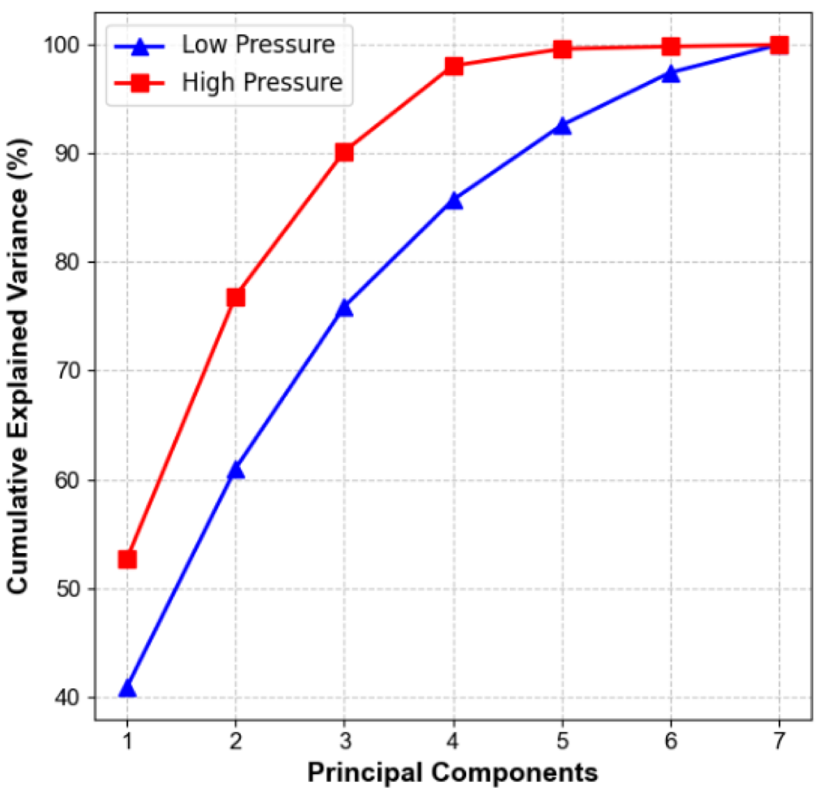

Figure 2—SVD analysis of input features for high- and low-pressure wells.

Figure 3 presents a comparative analysis of the feature contributions to the top 3 modes for both well types. For high-pressure wells, the first principal component is primarily driven by the contributions of Choke, THP, and Temperature DS. In contrast, the second mode is mainly influenced by wax-related properties (WAR, WCF) and Gas Lift Injected. At the low-pressure wells, the first mode reflects similarly strong contributions from Choke, THP, and, additionally, Pressure DS and Gas Lift Injected.

The second mode is mainly governed by WCF and WAR, highlighting the greater influence of the wax-related properties at low-pressure wells. These analyses indicate that surface and near-wellbore operating parameters, particularly Choke and THP, are the dominant contributors in the first singular mode across both pressure regimes. Meanwhile, wax-related properties take over in the second mode. The reduced number of modes required to explain the variance in high-pressure wells suggests more stable and predictable operational behavior. In contrast, low-pressure wells display a more complex, higher-dimensional response to interacting flow assurance and production control variables.

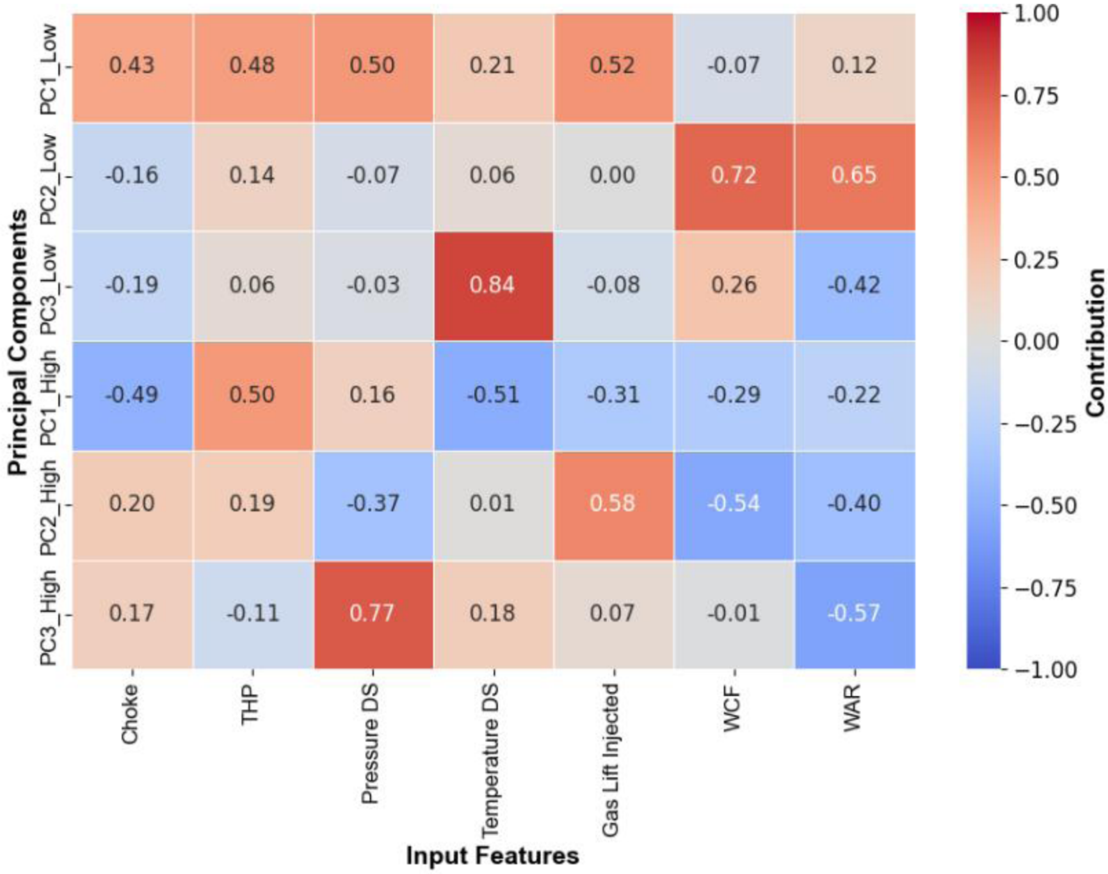

Figure 3—Relative contribution of inputs to the principal components for high- and low-pressure wells.

Due to the limited dataset (306 samples), data from both high- and low-pressure wells were combined, and a single ML model was developed for predicting production rates. The comprehensive comparison of the various ML models in estimating oil, water, and gas production rates is provided in Table 1. According to the results, linear regression consistently underperformed across all three targets, with R^2 values ranging from 0.41 to 0.46. MAE and RMSE values were also considerably higher than those of the other models, reflecting the limited capacity of the LR to capture complex, nonlinear relationships in production systems. In contrast, tree-based models reported superior performance in predicting production rates, with accuracy starting from R^2 scores of 0.93 for DT. Ensemble tree-based ML models (RF, ERT, GB, and CatBoost) achieved very high accuracies, reaching R^2 values of 0.98 to 0.99 for all three targets. MLP also produced reasonably accurate results with R^2 values of 0.84, 0.90, and 0.91 for oil, water, and gas, respectively. However, its performance was noticeably lower than that of the tree-based ensemble methods. This gap can be attributed to the relatively small dataset, which may not be sufficient for deep learning models to learn and generalize complex patterns effectively.

Table 1—Comparison of ML models based on the testing dataset with evaluation metrics

Model	MAE			RMSE			R^2		
	Oil	Water	Gas	Oil	Water	Gas	Oil	Water	Gas
LR	111.6	161.0	269.0	147.8	190.4	568.0	0.41	0.44	0.46
DT	23.1	28.9	52.3	39.7	51.1	118.5	0.95	0.93	0.96
RF	17.8	21.5	43.4	24.6	30.4	102.7	0.99	0.99	0.99
ERT	15.8	17.4	27.2	26.9	32.0	53.7	0.98	0.99	0.99
GB	19.8	24.8	36.1	26.4	31.8	68.5	0.98	0.99	0.99
CatBoost	19.6	25.7	35.3	29.4	36.2	67.3	0.98	0.98	0.99
MLP	49.6	60.0	179.2	82.8	88.9	273.6	0.84	0.90	0.91

It is worth highlighting that ERT delivered the best overall performance, with the lowest MAE (e.g., 15.8 bbl/day for oil, 27.2 Mcf/day for gas) and RMSE (e.g., 53.7 bbl/day for gas), confirming its robustness and effectiveness in handling limited but high-dimensional datasets. Figure 4 illustrates the comparison of the actual and predicted values of the training and test set target variables. Good alignment of ERT ML predictions and actual values indicates stable performance on training and testing data. It highlights the robustness and generalization capability of the tuned ML model for the task of production prediction.

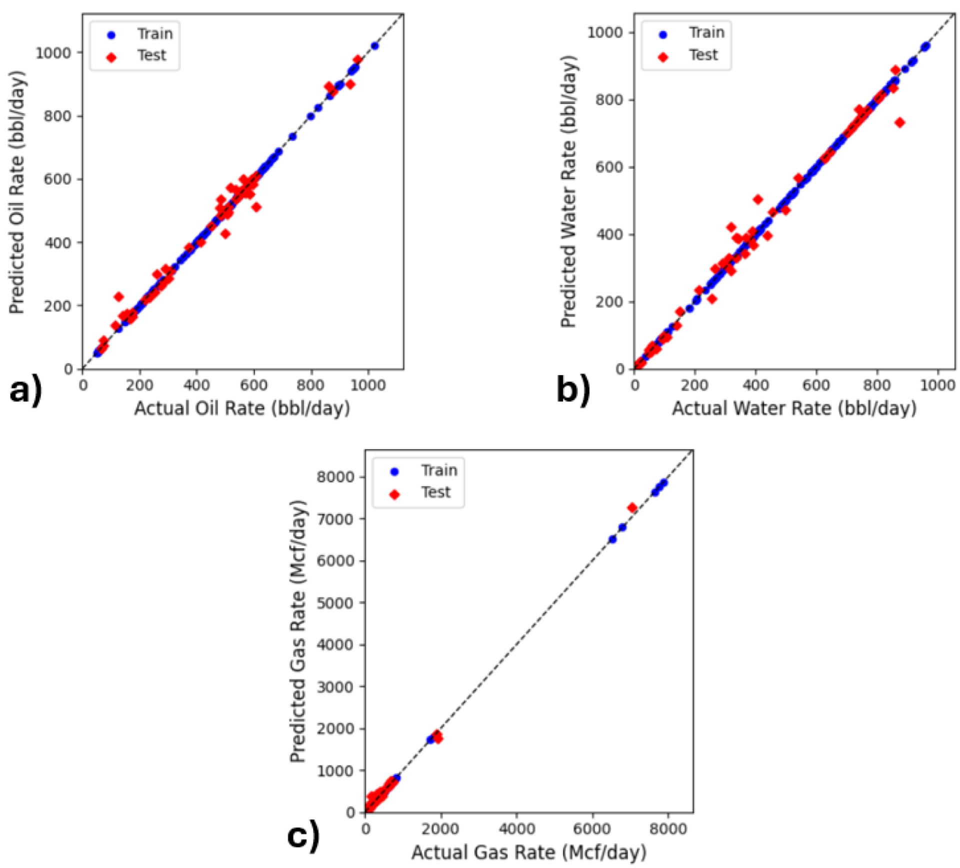

Figure 4—Comparison of predicted and actual values for train and test datasets of fluid production rate: a) Oil rate, b) Water rate, and c) Gas rate using ERT ML Algorithm.

To quantify the influence of flow assurance activities on well performance, a parametric sensitivity analysis was conducted using predictions from the optimized ERT model for both high- and low-pressure wells. These results revealed a complex relationship between the wax-related variables and the production rates of oil, water, and gas. The results, in Figures 5 and 6, demonstrate the detrimental impact of increasing both WAR and WCF on hydrocarbon production for both low-pressure and high-pressure wells, respectively. High-pressure wells had lower oil rates compared to low-pressure wells, which is more likely due to higher wax accumulation rate and more frequent interventions for its removal. On the other hand, high-pressure wells exhibited higher water and gas rates. Escalated WAR and WCF resulted in a decrease of up to 25% in oil production in low-pressure wells and 17% in high-pressure wells. In comparison, liquid production was reduced by up to 31% in low-pressure output and 11% in higher-pressure production. This production loss resulted from the cumulative effect of hydraulic restrictions due to wax deposition and wax-cut interventions, which caused downtime.

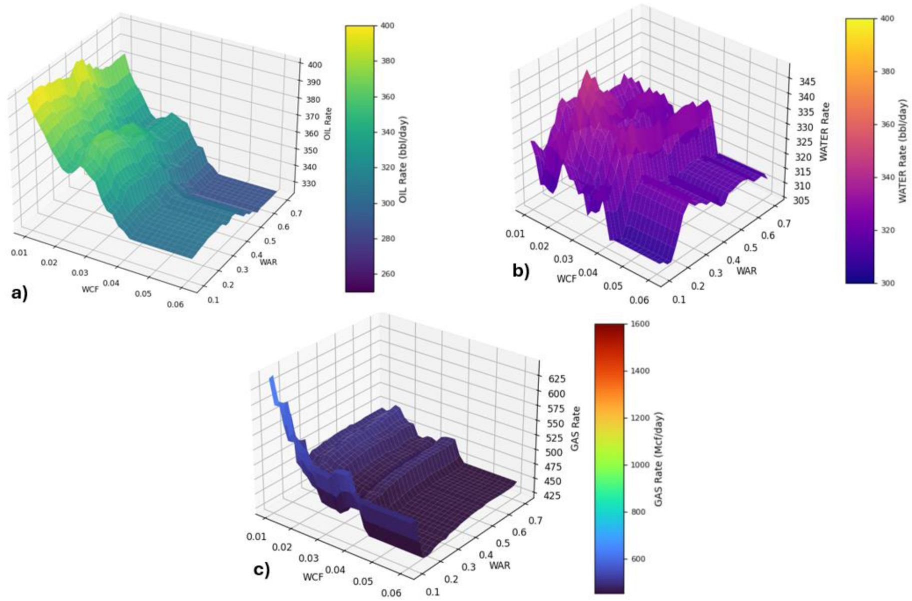

Figure 5—Fluid production rate as a function of WAR and WCF for Low-pressure well: a) Oil rate, b) Water rate, and c) Gas rate.

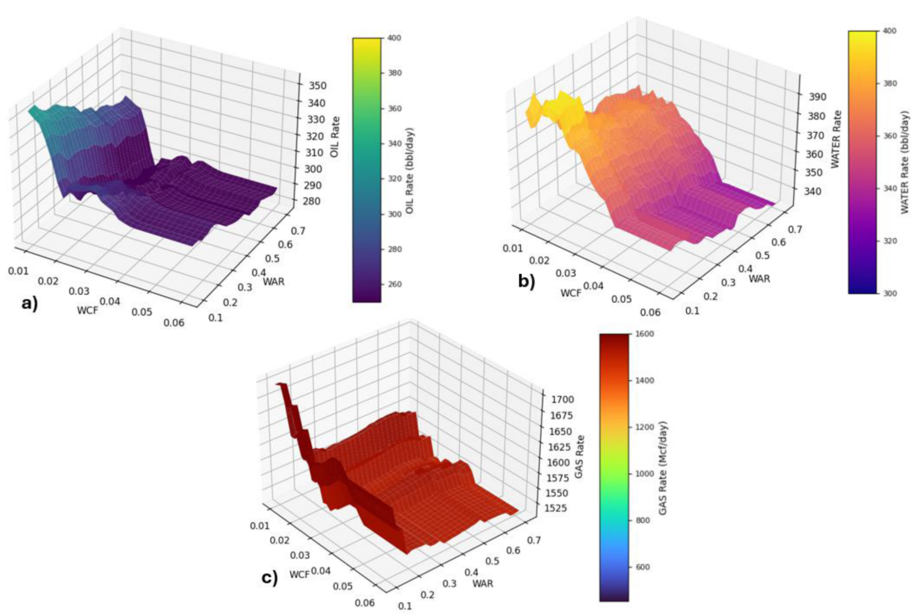

Figure 6—Fluid production rate as a function of WAR and WCF for High-pressure well: a) Oil rate, b) Water rate, c) Gas rate.

Analysis revealed varying reactions of each fluid phase to wax-related properties. Oil production was more sensitive to WCF, so that production rates significantly declined once the intervention frequency exceeded 0.01 h^{-1} (approximately one cut per 100 hours). This implies that the production downtime and flow instability resulting from frequent slickline operations can decrease the benefits of mechanical wax removal. On the other hand, WAR significantly attenuated gas production, and substantial output reductions occur when the accumulation rate exceeds 0.1 m/h. This is a sign of direct choking, physically caused by wax deposition, which lowers the effective hydraulic diameter, increases frictional pressure drop, and restricts general fluid flow in the borehole.

Increasing wax deposition also introduced more complex phase interactions; for instance, water production exhibited biphasic behavior. It initially increased with rising WAR and WCF before subsequently declining at higher levels of both parameters. This behavior is most likely an outcome of competing multiphase flow mechanisms. Wax deposition on the internal surface may change the roughness and wettability characteristics of the tubing. This would alter the liquid-liquid slip ratio between the oil and water phases, potentially enhancing water mobility and leading to a temporary increase in water cut. Meanwhile, the mobility of the oil and gas declined, which explains the initial drop in oil and gas rates as water levels rise. At intermediate values of WCF (0.025^{-1}h), there was a partial recovery of the oil and gas rates, accompanied by a decrease in the water rate, indicating a shift in mobility towards hydrocarbons. Further increases in wax accumulation and intervention rates led to a domination of gross hydraulic restriction. The narrowed internal diameter of the borehole resulted in a sharp increase in pressure losses, subsequently leading to an inevitable decline in the production rate of all fluids, including water. The reactions of oil, gas, and water rates to wax deposition were similar in high-pressure wells, as shown in Figure 6.

Collectively, these findings suggest that a reactive strategy for flow assurance through wax cutting can lead to significant production losses, as the oil rate is prone to frequent interventions. Proactive wax prevention, achieved through continuous chemical inhibition, may represent a more effective approach that can reduce the dependency on frequent mechanical cleaning. Maintaining a low WAR ensures stable flow conditions and avoids the production losses associated with both physical blockage and operational downtime. However, the choice of optimum strategy should ultimately be based on an economic study of the costs and benefits of chemical injection compared to mechanical wax cutting or a combination of both.

Summary and Conclusions

This work successfully demonstrates the potential of ML-based models as a tool for optimizing hydrocarbon production. By leveraging advanced ensemble learning algorithms, we developed an accurate model for oil, gas, and water rates. ML predictions were utilized to assess the impact of wax-related properties on the oil, gas, and water rates. The key findings are as follows:

- Ensemble tree-based models, particularly ERT, proved to be highly effective, delivering the most accurate predictions ($R^2 > 0.98$) for all three targets despite the dataset size limitations.
- WAR and WCF were found to be significant contributors to the well dynamics alongside THP and choke size, according to the SVD analysis.
- ML predictions for high-pressure wells showed lower oil rates compared to low-pressure ones, likely caused by wax deposition. This aligns with the input data, which indicates that high-pressure wells, on average, have twice the wax accumulation rate and a 50% higher intervention frequency compared to low-pressure wells.
- The sensitivity analysis revealed the detrimental effect of both WAR and WCF on production rates: oil and water rates were more sensitive to increasing WCF. In contrast, the gas rate was more severely affected by WAR. At elevated values of both wax-related properties, all flow rates decreased due to hydraulic restriction, underscoring the importance of careful monitoring when a reactive approach is employed.

- A proactive strategy that limits wax deposition through chemical inhibition could reduce reliance on frequent mechanical cutting, minimize operational downtime, and sustain production efficiency. Further studies are needed to evaluate the trade-off between chemical costs and intervention expenses, guiding optimal field implementation.

The primary limitation of this work is the scarcity of input data, which limits the model's resolution and the application of deep learning algorithms. Additionally, the absence of wax thickness to assess the magnitude of blockage also limits the prediction of the deposition's effect on production. Future work should focus on expanding the dataset with more frequent measurements, as well as incorporating additional features such as inhibitor injection rates, to account for the effect of proactive measures on the model. Integrating real-time sensor data and exploring hybrid ML models could further improve the accuracy of the predictions and their deployment for real-time, dynamic decision-making.

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Nomenclature

AOP = Asphaltene Onset Pressure (bar)
DS = Downstream
DT = Decision Tree
ERT = Extra Trees Regressor
GB = Gradient Boosting
MAE = Mean Absolute Error
ML = Machine Learning
MLP = Multilayer Perceptron
MSE = Mean Squared Error
RF = Random Forest
RMSE = Root Mean Squared Error
R^2 = Coefficient of Determination
SVD = Singular Value Decomposition
THP = Tubing Head Pressure (bar)
WAR = Wax Accumulation Rate (m/h)
WAT = Wax Appearance Temperature (°C)
WCF = Wax Cut Frequency (h^{-1})